

CASE STUDY 5: SYDNEY WATER CORPORATION – RISK-BASED APPROACH FOR LICENSING TO IMPROVE WET WEATHER OVERFLOW MANAGEMENT



Sydney Water and the NSW Environment Protection Authority (EPA) have co-created a refined approach to managing and regulating wet weather overflows, using a risk-based prioritisation methodology for Sydney's four major coastal sewage systems.

Management approach

Risk-based

Receiving environment principles considered/met/achieved

Ecosystem, Community

Background

In 1998, Sydney Water proposed frequency targets for wet weather overflows from its wastewater system. A key driver of this measure was to reduce the number of lost swimming days due to wet weather. Since then, the business has invested over \$1.5 billion to reduce the frequency of wet weather overflows. More than \$5.5 billion of investment would have been required to continue achieving the original frequency targets (2012), by building large storage tanks, tunnels and bigger pipes and pumps, resulting in a potential increase in customers' sewage bills with limited environmental or community benefit.

Sydney Water's area of operations has over 3,000 emergency relief structures (ERS) in highly urbanised areas. The geological features and topography in these areas limit the available options to store excess wastewater during wet weather, posing significant impact to the environment and community.

Since 2012, Sydney Water and the NSW Environment Protection Authority (NSW EPA) have been working on a risk-based approach in four major coastal wastewater systems to abating wet weather overflows (Northern and

Western suburbs – North Head EPL 378; Eastern suburbs – Bondi EPL 1688; Southern and Western suburbs – Malabar EPL 372; and Southern suburbs – Cronulla EPL 1728). This has been an iterative approach, with an initial simple prioritisation methodology to direct cost-effective investment in areas with known environmental and/or public health value during the 2020-24 Independent Pricing and Regulatory Tribunal (IPART) period. Further improvements since March 2020 have revised our methodology from a spreadsheet risk assessment to a data centric spatial prioritisation tool. The regulatory measure has also been changed to reflect a stronger focus on protecting the receiving environment.

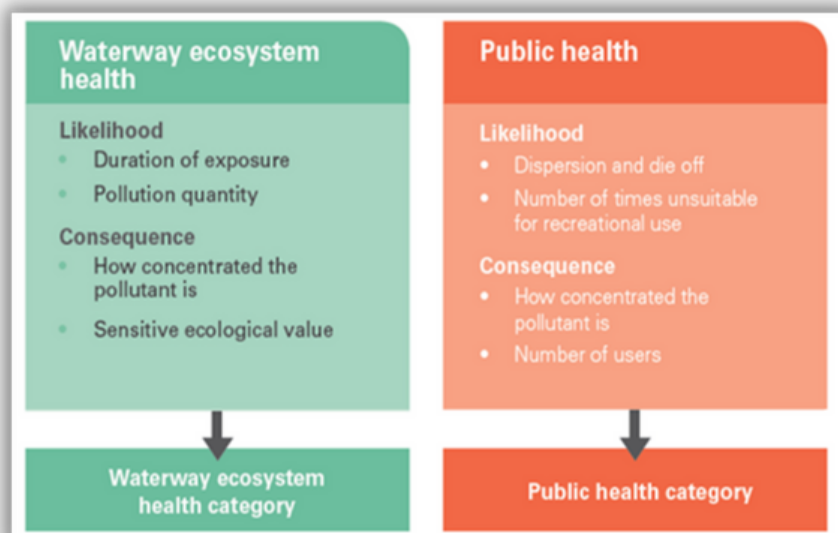
Investigations and program objectives

The risk-based prioritisation methodology incorporates community expectations regarding public health and the environment, the EPA's expectations on a new regulatory approach and Sydney Water's principal business objectives. It has been designed to be transparent, repeatable and allow for a focus on abating wet weather overflows in locations with the highest potential for impact.

The prioritisation uses a manual spreadsheet assessment which incorporates the best information available (at the time) to a system-wide reduction of volume and frequency, while considering the sensitivity and use of receiving waters (waterway ecosystem health, public health).

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FIGURE 1



Experts and stakeholders were consulted to develop the risk criteria and associated measures for each of the waterway values. Sydney Water assessed the current risk of overflow points using data from geographic information system layers and modelled wastewater system performance. The risk assessment results were then combined into a risk profile. This method was used to identify higher risk overflow points across the systems within the priority areas, and to inform the improvements compared to a baseline.

For the 2020–24 IPART period, the EPA is implementing a flexible regulatory point system (Pollution Reduction Program 307) to replace the previous frequency targets in the four major coastal wastewater systems (North Head, Bondi, Malabar & Cronulla). Abatement works are required on category 1 sites firstly (highest risk) and the majority of regulatory points must be achieved through this means. Some regulatory points can then also be achieved through abatement of category 2 and 3 sites under a sliding scale set of point ratios and rules. At the

end of the improvement period, it must be demonstrated that volume and/or frequency of overflow points has been reduced from the baseline risk profile. Each subsequent improvement period would incorporate an updated baseline risk profile. Through this process of adaptive management, we will reduce risks from wet weather overflows over time.

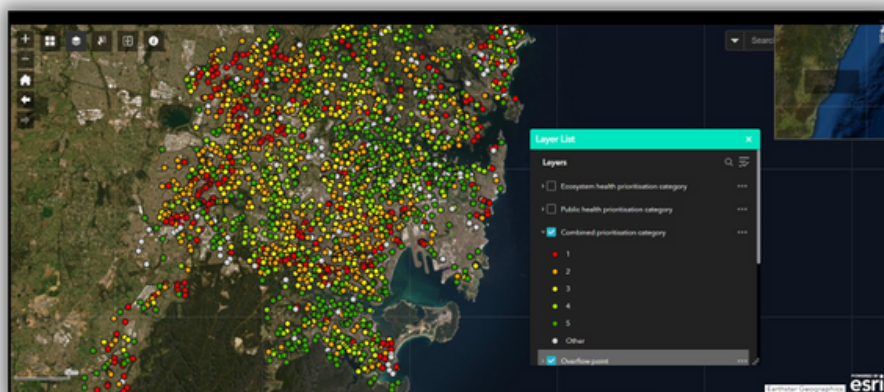
A recent revision of the prioritisation methodology will be applied to the next price period (2025–30). The revised prioritisation methodology relies on a data centric spatial model, (gathered from sources such as Sydney Water, NSW Government departments, councils and mobile phone usage) to improve the risk assessment allocation by using updated data and software to create a sophisticated and robust tool (Figure 2, Figure 3).

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FIGURE 2. Principles of the spatial prioritisation tool.



FIGURE 3. Preliminary risk-based prioritisation results from the spatial tool.



Sydney Water and the EPA agreed to an interim regulatory measure for 2024-25 of 1% total volume reduction of Category 1 ERS, from the four large coastal systems.

Stakeholder involvement

Sydney Water consulted customers, councils, community, industry partners and environmental groups on how they want to use waterways now and in the future. Guided by the IAP2 Public Participation Spectrum on Engagement, the community and stakeholder engagement strategy for the first iteration of the prioritisation methodology (2020-24) reflected the stakeholder feedback and their expectations on maintaining a healthy waterway.

As part of the engagement process, a range of workshops, site tours and online discussion forums were hosted to gauge community and stakeholder views on the proposal. Participants generally saw a benefit by using the risk-based approach to inform a new regulation and as a mechanism to reduce wastewater overflow. This valuable feedback has been used to test the thinking behind the proposal, identify gaps as the process develops, and refine how information is presented so stakeholders can understand the proposed approach.

The recent revision to the prioritisation methodology was developed in consultation with the EPA and the NSW Department of Planning and Environment (DPE).

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Outcomes

The risk-based prioritisation methodology to address wet weather overflows:

- better represents potential for impact on sensitive receptors
- can adapt to changing environments and community values
- provides a holistic approach that is fair and equitably applied across all areas
- effectively targets improvement where it is most needed
- allows integration and use of better data, tools and solutions to deliver environment and community benefits cost effectively
- is easy to access and use
- is capable of informing other projects by providing a framework that could be developed to assess risk for other assets and activities

Each ERS in Sydney's four large coastal wastewater systems was analysed as part of this project. The results prioritised the ERS with the highest potential for risk to the environment and community and highlighted where abatement work would deliver greater environment and community benefit. The regulation-based approach enables targeted and effective source control solutions to reduce the risk of overflows.

Costs/benefits

The Sydney Water risk-based prioritisation methodology and new regulatory measure was developed in two phases:

- \$4.9 million for the 2020-24 IPART period manual spreadsheet risk-based prioritisation and interim regulatory measure, which included:
 - a wet weather overflow strategy
 - the proposal to the EPA for a licence variation
 - the development of the prioritisation methodology, including the water quality models as an input into the prioritisation methodology
 - community and stakeholder engagement cost benefit valuation and analysis
 - costings for implementation

• \$1.1 million to revise risk-based prioritisation to a data centric spatial prioritisation tool applied to the 2024-30 IPART period. This included:

- sourcing suitable and relevant data as inputs
- building a spatial prioritisation tool
- engagement workshops with the EPA and DPE
- costing for implementation

The implementation of wet weather overflow abatement solutions is in addition to these costs.

Project timing

Development of a risk-based prioritisation methodology for wet weather regulation is an iterative process. Continuous improvement is key to ensure that the most relevant data is used to assess potential for risk. The Sydney Water's multi-disciplinary team undertook for the:

- 2020-24 IPART price period – approximately 6 months to develop the initial risk-based prioritisation model
- 2025-30 IPART price period – an additional 6 months to upgrade to a data centric spatial tool
- negotiations on the inputs and application of the revised prioritisation methodology and resultant regulatory measure – 12 to 18 months (for each iteration) of workshops with the EPA and DPE

The length of time may change and will depend on the resourcing, level of data accuracy/extent and regulator requirements.